STAFF REVIEW · H446 182 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. 3, 2, 1, 0. The n=0 call is the base case.

2. A call transfers control into a function; return finishes that invocation and gives its result to the caller.

3. Q. The topmost R is removed first.

4. Square(2)=4 and Square(3)=9; the total is 13.

I Do — source worked steps

1. Push — The frames are SumTo(3), SumTo(2), then SumTo(1), with SumTo(1) at the top.

2. Stop — SumTo(1) is the base case; it returns 1 first.

3. Unwind — SumTo(2)=2+1=3, followed by SumTo(3)=3+3=6.

We Do — guided CHECK

1. Top frame
SumTo(1) is on top immediately before the first return.
Returns
1, 3, 6, 10, 15. SumTo(5) returns 15.

You Do — source mark schemes

1. SumTo(4), SumTo(3), SumTo(2), SumTo(1). SumTo(1) is the top frame.

2. 1, 3, 6, 10, 15, 21 in the order calls return; the outer function returns 21.

3. The expression describes the final arithmetic but not execution order. The deepest call SumTo(1) must return first; only then can waiting calls compute their sums.

Exit Ticket — source answer

1. False. Each active call has its own stack frame and local values.